

SECTION AAnswer **all** questions.

1. (a) Radon is a radioactive gas and emits alpha particles.

(i) Explain what is meant by the *activity* of a radioactive source **and** give its unit. [2]

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(ii) The half-life of radon is 3.3×10^5 s. Determine the time it takes for a sample of radon to decay to 20 % of its initial activity. [3]

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(iii) Discuss the extent to which alpha particles are a risk to the human body. [3]

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